

Protecting Salem's Ecosystem

Predictive Modeling of Tree Canopies and Stream Temperatures

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Abstract

Salem's ecosystem is being threatened by rising stream temperatures due to climate change and the increasing prevalence of Emerald Ash Borers. The loss in tree canopy coverage and increasing temperatures risks pushing native species to cooler bodies of water and away from Salem's waterways. This issue must be addressed by the City of Salem to prevent loss of salmon population or other key species.



Introduction & Motivation

Problem & question:

Can we use historic stream temperature data to predict stream temperature for 2020, 2022, and 2024?
Can we combine historic stream and weather data to get a more accurate prediction of stream temperatures in 2020, 2022, and 2024?
Can we combine historic stream and canopy data to get a more accurate prediction of stream temperatures in 2020, 2022, and 2024?
Will combining all three historic datasets—streams, weather, and canopy—produce a highly accurate prediction of stream temperature in 2020, 2022, and 2024?

Impact & stakeholders:

The City of Salem has given us guidance in exchange for results that are actionable. In response to our findings, the city will likely implement preventative measures for stream temperature increases. This will impact indigenous communities who rely on Salmon populations, citizens who utilize streams for cooling or drinking, and anyone else who uses water in Salem, recreationally or otherwise. The risks to species as well as the potential for abundance of harmful algae could have serious impacts for Salem's population.



Background & Related Work

Bodies of water around the world are beginning to heat up at faster rates than air heat waves [1]. These water heat waves—defined by at least five consecutive days above the seasonal average—result in lower dissolved oxygen content, nutrient cycling, and pH buffering [3]. All of this means that ecosystems and native species adapted to live in colder waters face serious threats, while algae has room to become prolific.

One way to combat high stream temperatures is to support riparian zones, the dense plant areas along rivers [4]. These areas often sport trees and other foliage that shade and cool rivers and streams, effectively lowering the water temperature. However, these areas are at risk for several reasons.

This recent expansion of threats to river temperatures has caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan [2]. No research currently exists on stream temperatures, weather, and tree canopies in Salem, to our knowledge. This research will support government and private organizations in the prevention of the loss or disruption of riparian zones in Salem

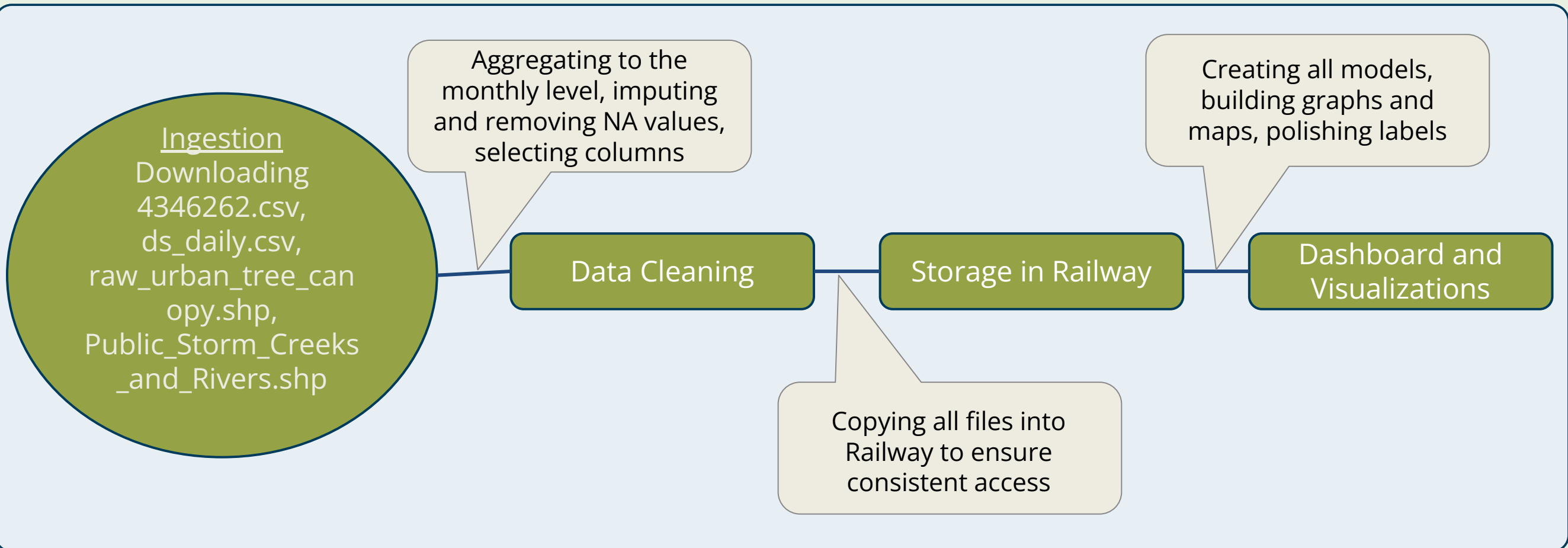


Data, Engineering & Ethics

Sources & scale: The data is sourced from NOAA, The City of Salem, Tree Plotter Canopy, and CastawayWU. These were all available with permission from the data publisher, and could be cleaning downloaded as a flat CSV file.

Ethics & responsible use: Our data contains no PII, so we are not concerned about accidentally sharing people's personal information. All of our data is publicly available, so we are not concerned about consent or redaction. Our main ethical concern with this project is creating predictions with the limited years of data that we have access to. This risks missing overall trends that are seen across years. This also limits our ability to make predictions into the future, rather we must focus on emphasising the scope of stream temperature change in Salem between years.

Pipeline:



Methods & Evaluation

Approach:

For each research question, multiple regression models were built for 2020, 2022, and 2024. The models include:

- Linear Regression
- Random Forest Regression
- SVR
- Elastic Net Regression
- Bayesian Ridge Regression
- XGB Regression
- CatBoost Regression
- Voting Regression

Validation:

These models were evaluated using the RMSE, MAE, R^2 , and 5-Fold Cross Validation.

The RMSE and MAE values were compared relatively between all models that were created. These metrics gave a sense of the error in the model's predictions. In regard to the R^2 value, a successful model was defined as having an R^2 value above 0.75. This provides a numeric representation of how well the model fits the variation in the data. Cross validation was used to determine how well the model would predict on unseen data.

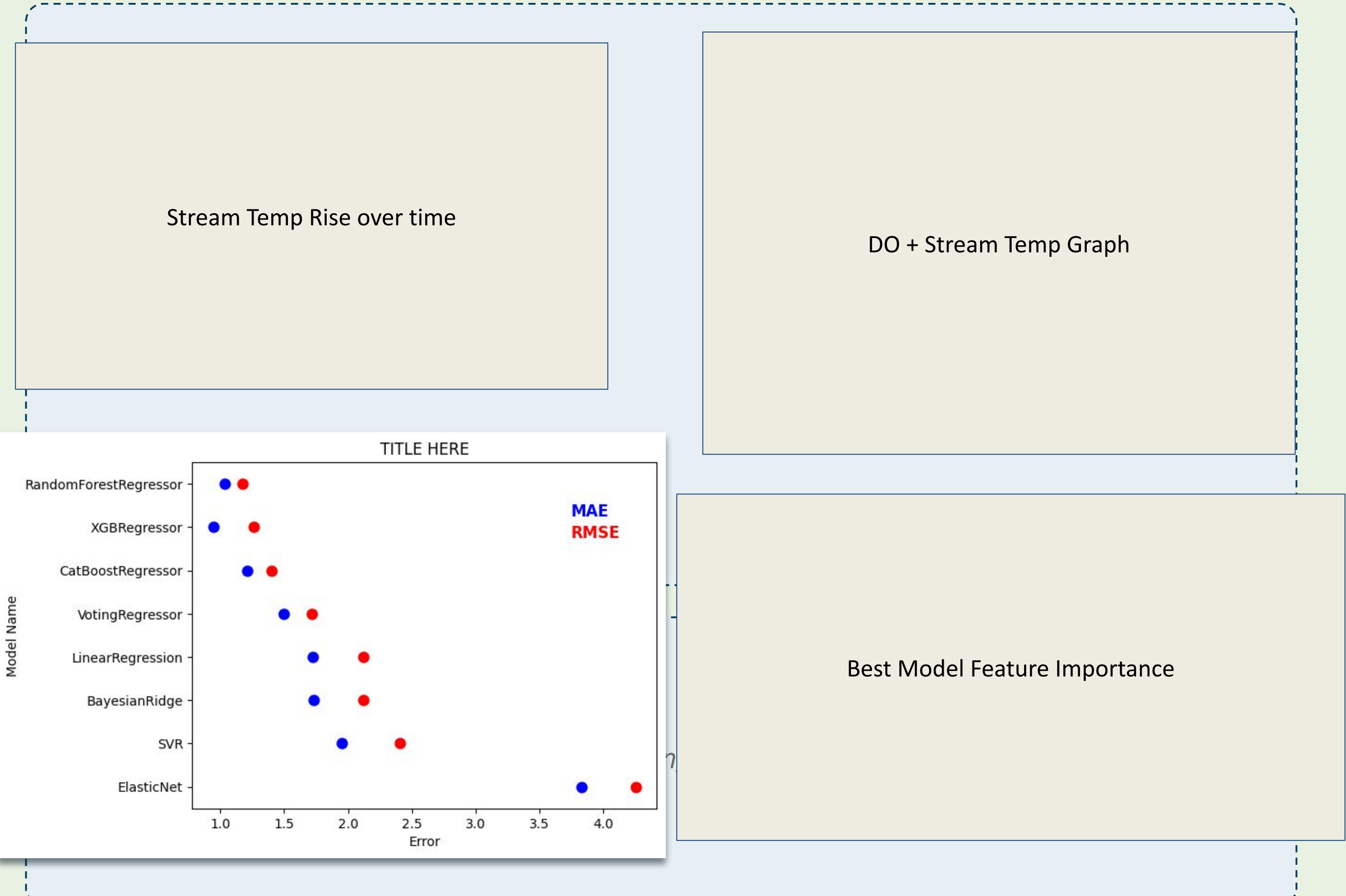


Results & Visualizations

Label every figure: axes, units, sample size (n).

Claims must match the evidence shown. Mark draft numbers as DRAFT.

We will include the Correlation between stream temp and tree canopy as well as the correlation between stream temperature and weather



Discussion, Limitations & Conclusions

Discussion:

Tree canopy does have an impact on stream temperature. If tree canopy is not protected across Salem's waterways, the risks include: salmon habitats suffering, Increase in algal blooms, overall stream ecosystem health

Limitations & future work:

Our data is extremely limited due to only having 3 years of data for tree canopy data.

In the future, we can use the models we created as a base to create a new model with more years of data, or adding more variables such as EAB infestation or human tree removal patterns.

Conclusions:

We can accurately predict stream temp within the year of the data. We have two main calls of action:

- Protecting tree canopy along waterways in Salem is important for maintaining healthy stream temperature
- Further data collection would greatly impact the ability to model risks to stream temperature.
 - It would be helpful to have tree canopy data for a wider range of years or data on the number of trees and the types of trees that exist across different years. A great way to do this would be LiDAR data for each year in the future.



Reproducibility & References

Repository: <https://github.com/Serenna4/Emerald-Ash-Borer-Capstone-Project/tree/main>

Key dependencies: Python 3.11.1, R 4.5.2, Beekeeper SQL 5.9, Railway | Libraries: NumPy, Matplotlib, Seaborn, Pandas, SciKit Learn

References: [1] Bush, E., U.S. rivers are experiencing unprecedented and unexpectedly intense warming, NBC News, 2025.

[2] City of Salem, City of Salem Emerald Ash Borer Readiness and Response Plan, City of Salem, 2026.

[3] Penn State Extension, We are in Hot Water—Streams and Rivers are Heating Up, Pennsylvania State University, 2026

[4] University of Georgia, Stream Warming and Brook Trout in the Southeast, University of Georgia, 2026.

Acknowledgements

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